0 to — ← unsigned
-ve to +ve ← signed

| short ✓
| long ✓

Data types :-

1. Primary → int
   1. (signed) int
   2. (signed) short int
   3. (signed) long int
   4. unsigned int
   5. unsigned short int
   6. unsigned long int
2. Derived
3. user-defined

16 bit
32 bit ✓
64 bit

int x;

100

x

x = 7.22; ⇒ x = 7;

x = 100000000000000000000000000000000;

unsigned int no;

no = 72; ✓ || no = -5; ✗

int num; || signed int num;

num = -7321; ✓

num = 5923; ✓

* Primary Datatypes :-

1) int
2) void
3) char → (signed) char, unsigned char
4) floating point
   - float
   - double
   - long double

→ char

d

myChar

char myChar;

myChar = 'A'; // myChar = 65;

// myChar = 'd';

// myChar = 97;

65

10

1000001

1001110

ASCII value

A → 65
a → 97
(zero) 0 → 48

B = 66, C = 67, D = 68 - - - -
b = 98, c = 99, d = 100 - - - - -

format specifier → %c

```
scanf("%c", &myChar);
printf("%c", myChar);        // A

printf("%c %d", myChar, myChar); // A 65
```